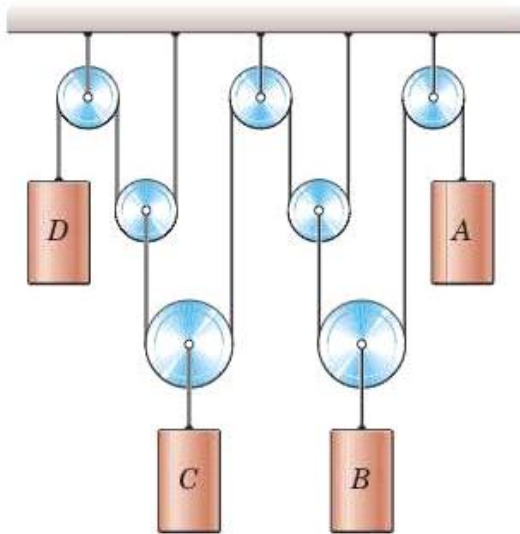
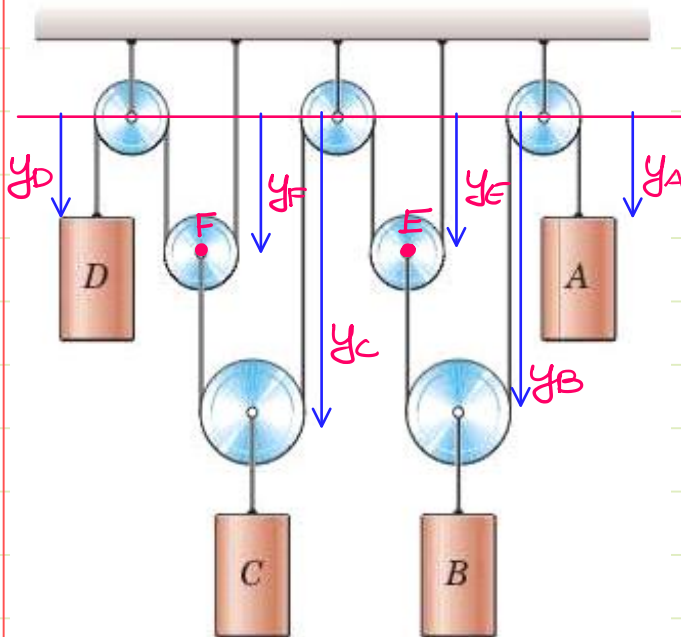




$$\begin{aligned} v_A &= 2 \text{ m/s} \downarrow \\ v_B &= 3 \text{ m/s} \uparrow \\ v_C &= 1 \text{ m/s} \downarrow \\ v_D &= ? \end{aligned}$$



$$\begin{aligned} y_A + y_B + y_E - y_F &= l_1 \\ \Rightarrow y_A + 2y_B - y_E &= l_1 \quad (1) \end{aligned}$$

$$2y_E + 2y_C - y_F = l_2 \quad (2)$$

$$2y_F + y_D = l_3 \quad (3)$$

$$\begin{aligned} \text{by } (2) \text{ \& } (3) \\ 2\{2y_E + 2y_C - l_2\} + y_D &= l_3 \\ \Rightarrow 4y_E + 4y_C + y_D &= C \end{aligned}$$

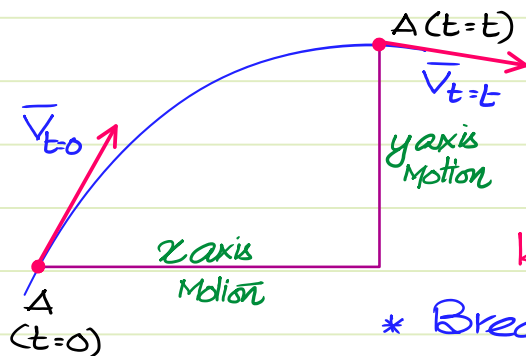
$$\text{from } (1) : y_E = y_A + 2y_B - l_1$$

$$\begin{aligned} \Rightarrow 4\{y_A + 2y_B\} + 4y_C + y_D &= C \\ \Rightarrow 4y_A + 8y_B + 4y_C + y_D &= C \\ \Rightarrow 4\dot{y}_A + 8\dot{y}_B + 4\dot{y}_C + \dot{y}_D &= 0 \end{aligned}$$

$$\therefore 4(2) + 8(-3) + 4(1) + \dot{y}_D = 0 \quad \therefore \dot{y}_D = 12 \text{ m/s}$$

$$\therefore v_D = 12 \text{ m/s} \downarrow$$

3.1.2. Curvilinear Motion



- * Multiple translatory DOF.
- * Since the velocity is always tang. to the pathline, $\therefore$ the dirⁿ of $\vec{V}$ keeps on changing

* Breakdown the $\vec{V}$ into mutually $\perp$ axes. If Motion is analysed separately in these axes.

Note: The motion in the $\perp$ axes is related to each other by time $\Rightarrow$ Time taken for motion in every axis is same.

* Eq^{ns} of Motion:

x-Axis Motion:

1. Uniform accⁿ

$$V_x = U_x + a_x t$$

$$x = U_x t + \frac{1}{2} a_x t^2$$

$$V_x^2 - U_x^2 = 2a_x x$$

2. Non Uniform accⁿ

$$V_x = dx/dt$$

$$a_x = dv_x/dt$$

$$V_x \cdot dv_x = a_x \cdot dx$$

Similar set of eq^{ns} appear if the motion has y axis or z axis.

Solution Eqⁿ of traj is parabolic $\therefore$ Curv. M.

$$y = x^2 + 4x + 100 \quad \{y = f(x)\}$$

$$\vec{U} = \{4\hat{i} + 16\hat{j}\} \text{ m/s.}$$

$$a_x = 0 \Rightarrow V_x = 4 \text{ m/s.}$$

07. A Particle moves along the path $y = x^2 + 4x + 100$, starting with an initial velocity of $\vec{V}_0 = (4\hat{i} + 16\hat{j})$ m/s. If V_x is constant, the value of V_y and a_y at $x = 16$ m will be

(a) $V_y = 100$; $a_y = 35$

(b) $V_y = 144$; $a_y = 32$

(c) $V_y = 90$; $a_y = 40$

(d) $V_y = 110$; $a_y = 45$

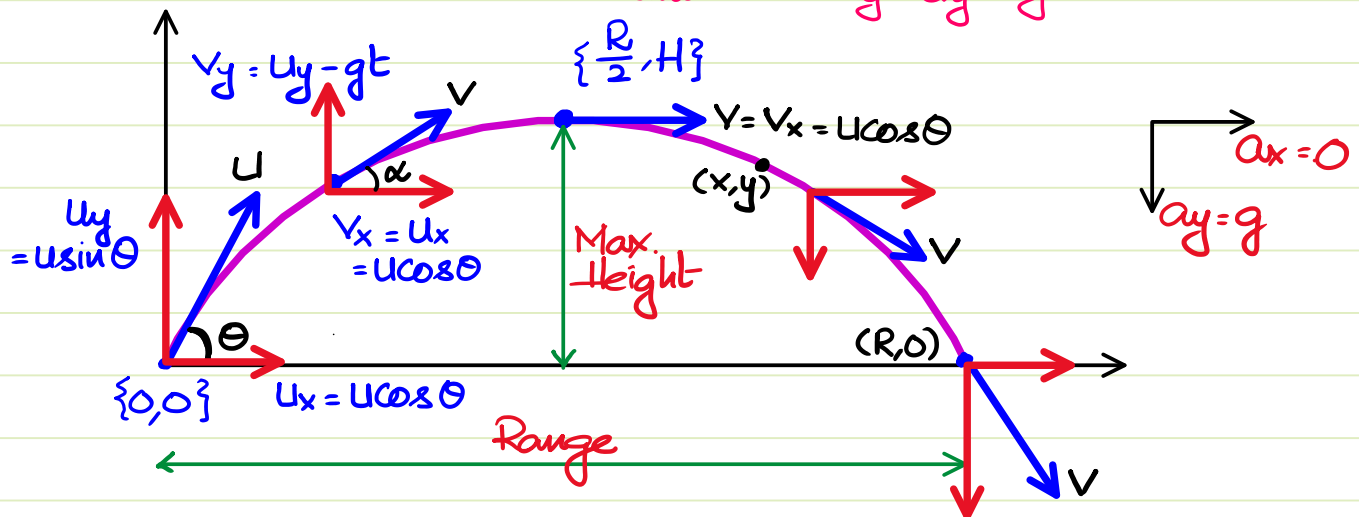
$$V_y = \frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} \quad \text{Since } V_x = 4 \quad \therefore V_y = 4(2x + 4) = 8x + 16 \quad \therefore V_y = 144 \text{ m/s. at } x = 16 \text{ m}$$

$$a_y = \frac{dV_y}{dt} = \frac{dV_y}{dx} \cdot \frac{dx}{dt} \quad \text{Since } V_x = 4 \quad \Rightarrow a_y = 32 \text{ m/s}^2$$

Projectile Motion: Motion of a body projected at a velocity at a particular angle in a given accⁿ field. {We assume Gravitational field}

$$\{a_x = 0 \text{ m/s}^2 ; a_y = -g \text{ m/s}^2\}$$

$$\Rightarrow V_x = \text{constant} \quad \Rightarrow V_y = U_y - gt$$



U : Initial velocity ; θ : Angle of projection.

1. Eqⁿ of trajectory: Rel. ship b/w x & y co-ordinates along trajectory.

Motion in x -axis: $x = U_x t + \frac{1}{2} a_x t^2 \Rightarrow \boxed{t = \frac{x}{U_x}}$

Motion in y -axis: $y = U_y t + \frac{1}{2} a_y t^2$
 $\Rightarrow y = U_y \left\{ \frac{x}{U_x} \right\} + \frac{1}{2} (-g) \left(\frac{x}{U_x} \right)^2$

$$\Rightarrow y = \tan \theta \cdot x - \frac{g x^2}{2 U^2 \cos^2 \theta}$$

$$\Rightarrow \boxed{y = \left\{ -\frac{g \sec^2 \theta}{2 U^2} \right\} x^2 + \tan \theta \cdot x} \quad \therefore \text{traj is parabolic in nature.}$$

Max. Height: $\boxed{H = \frac{U^2 \sin^2 \theta}{2g}}$

Range: $\boxed{R = \frac{U^2 \sin 2\theta}{g}}$

Time of flight: $\boxed{T = 2U \sin \theta / g}$

Given: $U = 20 \text{ m/s}$

$$x = 20 \text{ m}, y = 8 \text{ m}$$

To find: θ

$$\text{Eqn: } y = \left\{ -g \frac{\sec^2 \theta}{2u^2} \right\} x^2 + \tan \theta \cdot x$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$\Rightarrow 8 = \left\{ \frac{-9.81(1 + \tan^2 \theta)}{2 \times 20^2} \right\} 20^2 + \{\tan \theta\} 20$$

$$\Rightarrow 4.905 \tan^2 \theta - 20 \tan \theta + 12.905 = 0$$

$$\Rightarrow \tan \theta = 3.27, 0.8$$

$$\theta = 73.08^\circ, 38.47^\circ$$

11. A particle is projected with a velocity of 20 m/s in air at an angle of θ with horizontal. The x and y coordinates of a point lying on the trajectory of the particle with respect to point of projection are 20 m and 8 m respectively. The angle of projection θ is equal to:

(a) 73.08° or 38.47°

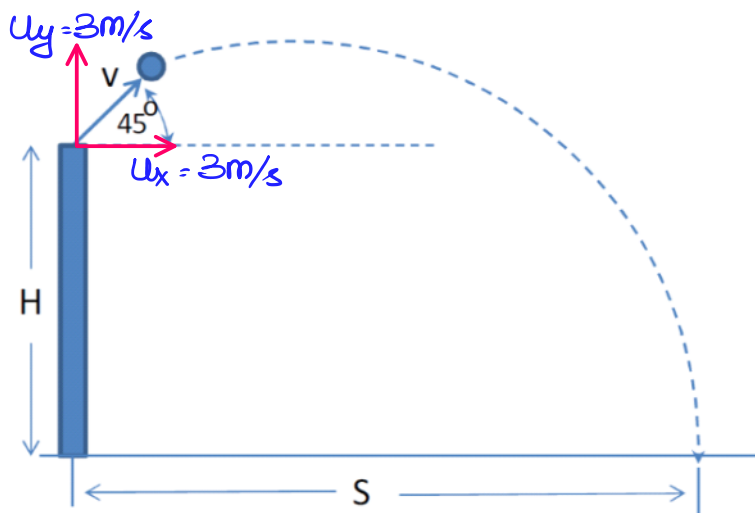
(b) 77.08° or 40.47°

(c) 80.08° or 44.47°

(d) 83.08° or 48.47°

2M x 5

A projectile is fired with a velocity $V = 3\sqrt{2} \text{ m/s}$ from a point at height $H = 0.8 \text{ m}$ at an angle of 45° with respect to the horizontal direction as shown in the Figure. Find the horizontal distance s in meters travelled by the projectile when it hits the ground. Take acceleration due to gravity as 10 m/s^2



for Motion in y axis:

$$y = -0.8 \text{ m}, U_y = 3 \text{ m/s}$$

$$a_y = -10 \text{ m/s}^2$$

$$\Rightarrow y = U_y t + \frac{1}{2} a_y t^2$$

$$\Rightarrow -0.8 = 3t - 5t^2$$

$$\Rightarrow 5t^2 - 3t - 0.8 = 0$$

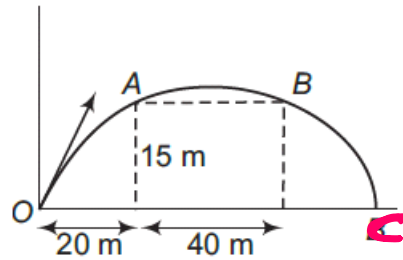
$$t = 0.8 \text{ s} \text{ \& } -0.2 \text{ s.}$$

for Motion in x axis: $x = U_x t + \frac{1}{2} a_x t^2$

$$x = 3 \times 0.8$$

$$= 2.4 \text{ m.}$$

2M. In the projectile motion shown in figure, given $t_{AB} = 2$ s then ($g = 10 \text{ ms}^{-2}$)



- ☒ (a) particle is at point B at 3 s
- ☒ (b) maximum height of projectile is 20 m
- ☒ (c) initial vertical component of velocity is 20 ms^{-1}
- ☒ (d) horizontal component of velocity is 20 ms^{-1}

Analysing AB: $x_{AB} = u_x \cdot t_{AB}$
 $\Rightarrow 40 = u_x \{2\}$
 $\Rightarrow u_x = 20 \text{ m/s} \quad \{d\}$

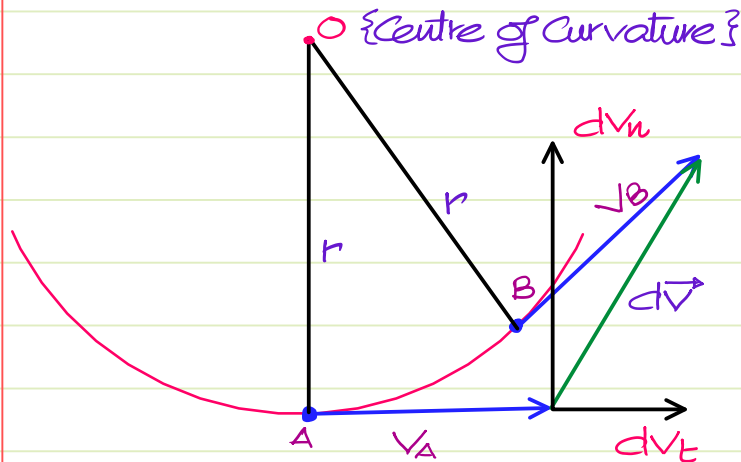
Analysing OB: $x_{OB} = u_x \cdot t_{OB}$
 $60 = 20 \{t_{OB}\}$
 $\Rightarrow t_{OB} = 3 \text{ s} \quad \{a\}$

Analysing OA: $y_{OA} = u_y t + \frac{1}{2} a_y t^2$
 $15 = u_y (1) - 5(1)^2$
 $\Rightarrow u_y = 20 \text{ m/s} \quad \{c\}$

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g} = \frac{u_y^2}{2g} = \frac{20^2}{20} = 20 \text{ m} \quad \{b\}$$

* Difference b/w Rectilinear Motion & Curvilinear Motion :

Velocity Vector is always tangential to the pathline regardless of Rectilinear or Curvilinear Motion.



In curvilinear Motion there are 2 components in change in Velocity.
{ dv_T , dv_N }

dv_T : changes the Mag. of Velocity.

dv_N : changing the dirⁿ of Velocity.

If $dv_N = 0 \Rightarrow$ Rectilinear Motion

Rectilinear Mot

$$dv_N = 0$$

$$\Rightarrow a_N = 0$$

only tang. accⁿ can exist

$$\left\{ \frac{v^2}{r} \right\} = 0$$

$$r \rightarrow \infty$$

$\vec{v}$ & $\vec{a}$ are collinear vectors

Curv. Motion

$$dv_N \neq 0$$

$$\Rightarrow a_N \neq 0$$

$$\vec{a} = \vec{a}_{\text{tang}} + \vec{a}_{\text{normal}}$$

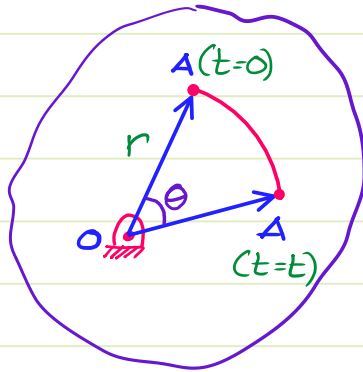
$$\frac{v^2}{r} \neq 0$$

$$r = \text{finite}$$

$\vec{v}$ & $\vec{a}$ are Non collinear vectors.

3.2. Kinematics of Rigid Bodies

3.2.2. Fixed Axis Rotation



* only 1 Rotational D.O.F.

Angular displacement $\{\theta\}$:

The change in direction of a radius vector (or) position vector of a body w.r.t. a reference. {S.I. : rad.}

Angular Velocity $\{\omega = \frac{d\theta}{dt}\}$: Rate of change of direction (or) angular displacement {SI: rad/s}

Angular Acceleration $\{\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}\}$: Rate of change of angular velocity {SI: rad/s²}

Equations of Motion in FAR:

A. Uniform Acc. Motion $\{\alpha = c\}$

1. $\omega = \omega_0 + \alpha t$
2. $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$
3. $\omega^2 - \omega_0^2 = 2\alpha\theta$

B. Non-Uniform Acc. Motion $\{\alpha \neq c\}$

1. $\omega = \frac{d\theta}{dt}$
2. $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$
3. $\omega \cdot d\omega = \alpha \cdot d\theta$

IM.

A wheel is spinning at 27 rad/s but is slowing with an acceleration that has a magnitude given by $3t^2$, in rad/s^2 for t in seconds. It stops in a time of

- (a) 1.7 s (b) 2.6 s
 ✓ (c) 3.0 s (d) 4.4 s

$$\omega_0 = 27 \text{ rad/s}$$

$$\alpha = -3t^2$$

$$\omega = 0 \text{ rad/s} \quad t = ?$$

$$\alpha = \frac{d\omega}{dt} = -3t^2$$

$$\Rightarrow \int_{27}^0 d\omega = - \int_0^t 3t^2 dt$$

$$\Rightarrow 27 = t^3 \Rightarrow t = 3 \text{ s}$$

IM.

The rotation of a fly wheel is governed by the equation $\omega = 4\sqrt{t}$; ω is in radians/sec and t is in seconds. If $\theta = 2$ radians when $t = 1$ s, find the values of θ and α at the instant when $t = 3$ s

- (a) $\theta = 14$ rad; $\alpha = 1.2 \text{ rad/s}^2$
 (b) $\theta = 15.2$ rad; $\alpha = 1.3 \text{ rad/s}^2$
 ✓ (c) $\theta = 13.2$ rad; $\alpha = 1.154 \text{ rad/s}^2$
 (d) $\theta = 14.5$ rad; $\alpha = 1.75 \text{ rad/s}^2$

$$\omega = 4t^{1/2}$$

$$t = 1 \text{ s}, \theta = 2 \text{ radian}$$

$$t = 3 \text{ s}, \theta = ? \text{ \& } \alpha = ?$$

$$\alpha = \frac{d\omega}{dt} = 2t^{-1/2}$$

at $t = 3 \text{ s}$.

$$\alpha = 2/\sqrt{3}$$

$$\alpha = 1.154 \text{ rad/s}^2$$

$$\omega = \frac{d\theta}{dt}$$

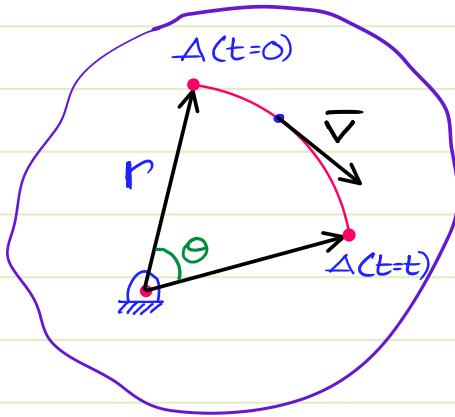
$$\Rightarrow 4t^{1/2} = \frac{d\theta}{dt} \Rightarrow \int_2^\theta d\theta = \int_1^3 4t^{1/2} dt$$

$$\theta - 2 = \frac{8}{3} \{3^{3/2} - 1\}$$

$$\Rightarrow \theta = 2 + \frac{8}{3} (3\sqrt{3} - 1)$$

$$\theta = 13.2 \text{ radian.}$$

* Conversion of Angular Parameters Into Linear Parameters :



Path travelled : $s = r \cdot \theta$

Instantaneous Velocity : $v = r\omega$

The $\vec{v}$ is always tangential to the path taken $\{ \perp^{\text{or}}$ to the radius vector & in the dirⁿ of $\omega \}$.

* Linear Acceleration : $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} (|\vec{v}| \cdot \hat{v})$

$$\vec{a} = |\vec{v}| \cdot \frac{d\hat{v}}{dt} + \hat{v} \cdot \frac{d|\vec{v}|}{dt}$$

Centripetal / Radial / Normal Accⁿ.
{changes the dirⁿ of $\vec{v}$ }

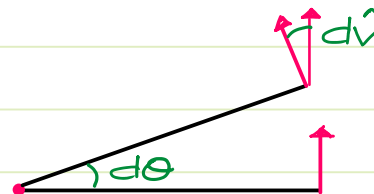
Tangential Accⁿ.
{changes the Mag of $\vec{v}$ }

* Centripetal Accⁿ :

$$a_c = |\vec{v}| \cdot \frac{d\hat{v}}{dt} = |\vec{v}| \cdot \frac{d\theta}{dt}$$

$$\Rightarrow a_c = r \cdot \omega \cdot \omega$$

$$\Rightarrow a_c = r\omega^2 = \frac{v^2}{r}$$



By similarity of Δ^{les}
 $d\theta = d\hat{v}$

Centripetal Accⁿ is directed along the radius & towards the centre of curvature {radially Inward.}

* Tangential Accⁿ : $a_T = \hat{v} \cdot \frac{d|\vec{v}|}{dt} = \hat{v} \cdot \frac{d(r\omega)}{dt}$ { $\hat{v}$ sep. that a_T is collinear to $\vec{v}$ }

$$\text{If } r = \text{const} \Rightarrow a_T = r \frac{d\omega}{dt} \Rightarrow \boxed{a_T = r \cdot \alpha}$$

a_T is always $\perp^{\text{an}}$ to r & in the direction of α .

$$\Rightarrow a_c \perp^{\text{ar}} a_T \quad \therefore \boxed{a_{\text{total}} = \sqrt{a_c^2 + a_t^2}}$$

Note: If $r \neq \text{const}$ in $a_t = \frac{d(r\omega)}{dt}$, then an additional tangential accⁿ known as Coriolis Accⁿ is developed.

1M.

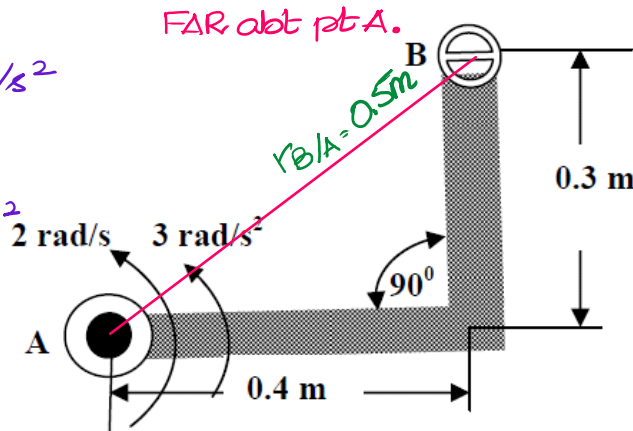
A "L" shaped robotic arm AB is connected to a motor at end A and a magnetic gripper at B as shown in the figure. If the arm is rotating with an angular velocity of 2 rad/s and an angular acceleration of 3 rad/s², the magnitude of the acceleration (in m/s²) of the end B is

$$a_c = r\omega^2 = 0.5 \times 2^2 = 2 \text{ m/s}^2$$

$$a_t = r\alpha = 0.5 \times 3 = 1.5 \text{ m/s}^2$$

$$a = \sqrt{a_c^2 + a_t^2}$$

$$\boxed{a = 2.5 \text{ m/s}^2}$$



(A) 2.0

☒ (B) 2.5

(C) 5.0

(D) 6.0

2M.

A particle P is moving on a circular path of radius $r = 1 \text{ m}$. The angular location θ of the particle is measured as shown in the Figure. The motion of the particle is described by $\theta = 2 \sin(t)$. Find the magnitude of the total acceleration (in m/s²) of the particle at time $t = \pi/3$ seconds.

$$\omega = \frac{d\theta}{dt} = 2 \cos t$$

$$\text{@ } t = \pi/3$$

$$\boxed{\omega = 1 \text{ rad/s}}$$

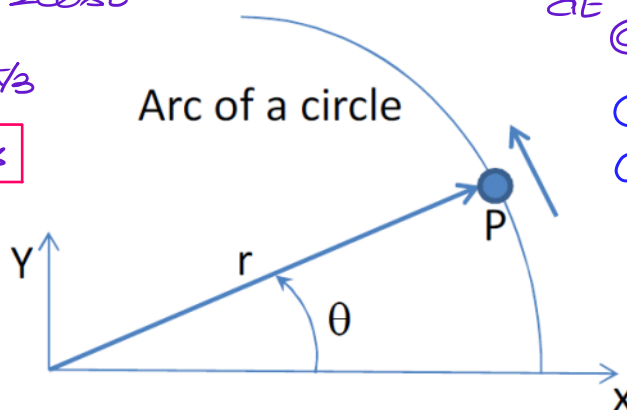
$$\alpha = \frac{d\omega}{dt} = -2 \sin t$$

$$\text{@ } t = \pi/3; \quad \boxed{\alpha = -\sqrt{3} \text{ rad/s}^2}$$

$$a_t = r\alpha = \sqrt{3} \text{ m/s}^2$$

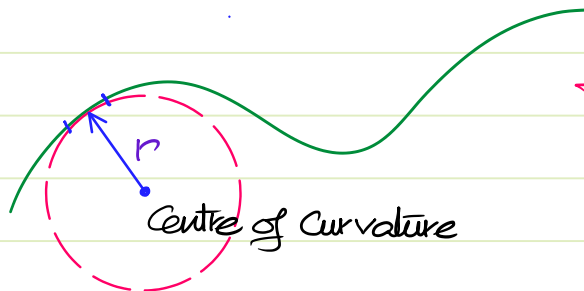
$$a_n = r\omega^2 = 1 \text{ m/s}^2$$

$$\boxed{a = 2 \text{ m/s}^2}$$



Radius of Curvature In Curvilinear Motion :

Any curved path can be imagined as assembly of elementary arcs of different circles. The radius of the circle whose arc is present on the curve is known as radius of curvature.



r : Radius of Curvature

To identify " r " use the concept of Centripetal Accⁿ:

$$a_c = \frac{V^2}{r} \Rightarrow \boxed{r = \frac{V^2}{a_c}}$$

V : Abs Velocity

a_c : Centripetal Accⁿ {component of Net accⁿ in the dirⁿ $\perp^u$ to $\vec{V}$ }

IM. The motion given is projectile Motion $\Rightarrow$ Abs. accⁿ = g

15. A particle moves up in the air with an initial velocity of 50m/s making an angle of 30° with horizontal. The radius of curvature at the top of its path is:
- (a) 175.25 m (b) 185.35 m
(c) 188.50 m (d) 191.12 m

Velocity at the top of trajec.:

$$V_x = U_x = U \cos \theta$$

$$= 50 \cos 30^\circ = 25\sqrt{3} \text{ m/s.}$$

Since the abs. velocity is in x axis, therefore { a_n } must be the component of abs. accⁿ { g } in y axis.

$$\because g \text{ is in } y \text{ axis} \quad \therefore a_n = g.$$

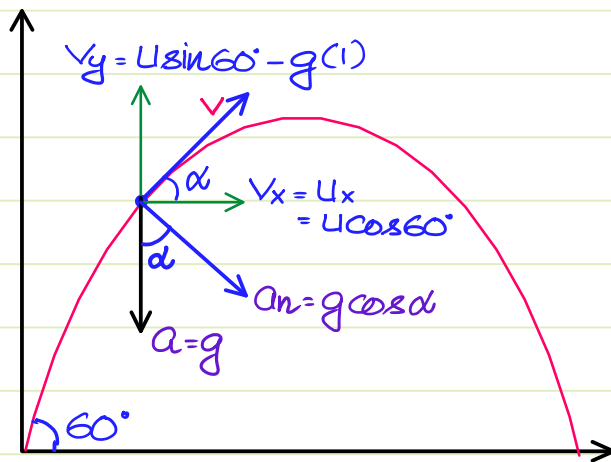
$$r = \frac{V^2}{a_n} = \frac{\{25\sqrt{3}\}^2}{9.81} \quad \therefore \boxed{r = 191.13 \text{ m}}$$

* The angle of projection wrt horizontal at which the radius of curvature at the top of path is equal to the Max height of trajectory is 54°

a. 36° b. 45° c. 54° d. 60°

$$r = \frac{V^2}{a_n} = \frac{U^2 \cos^2 \theta}{g} \quad ; \quad H = \frac{U^2 \sin^2 \theta}{2g}$$

$$\because r = H \quad \therefore \frac{U^2 \cos^2 \theta}{g} = \frac{U^2 \sin^2 \theta}{2g} \Rightarrow \tan \theta = \sqrt{2} \quad \theta = \tan^{-1} \sqrt{2}$$



14. A particle moves up in the air with an initial velocity of 100m/s making an angle of 60° with horizontal. The radius of curvature at time $t = 1$ sec is:
- (a) 1568.65 m (b) 1575.25 m
(c) 1625.50 m (d) 1650.75 m

$$r = \frac{V^2}{a_n} = \frac{\{V_x^2 + V_y^2\}}{g \cdot \cos \alpha}$$

$$\cos \alpha = \frac{V_x}{V} = \frac{V_x}{\{V_x^2 + V_y^2\}^{1/2}}$$

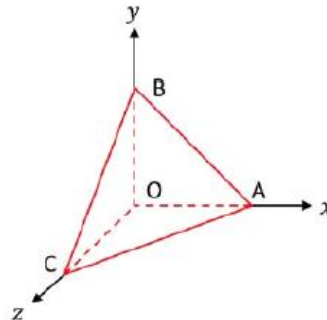
$$\Rightarrow r = \frac{(V_x^2 + V_y^2)^{3/2}}{g \cdot V_x}$$

$$V_x = U_x = 100 \cos 60^\circ = 50 \text{ m/s}$$

$$V_y = U \sin \theta - gt = 100 \sin 60^\circ - 9.81 = 76.7925 \text{ m/s}$$

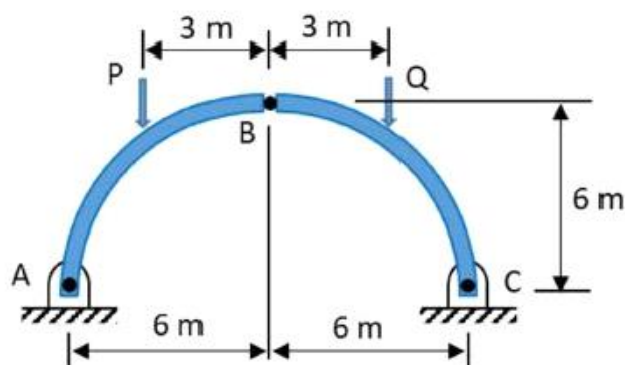
$$\Rightarrow r = 1568.65 \text{ m}$$

- Q.21 A rigid massless tetrahedron is placed such that vertex O is at the origin and the other three vertices A, B, and C lie on the coordinate axes as shown in the figure. The body is acted on by three point loads, of which one is acting at A along x-axis and another at point B along y-axis. For the body to be in equilibrium, the third point load acting at point O must be



- Options
- ☒ A. in $x - y$ plane but not along x or y axis
 - ☐ B. along z -axis
 - ☐ C. in $y - z$ plane but not along y or z axis
 - ☐ D. in $z - x$ plane but not along z or x axis

- Q.41 A three-hinge arch ABC in the form of a semi-circle is shown in the figure. The arch is in static equilibrium under vertical loads of $P = 100$ kN and $Q = 50$ kN. Neglect friction at all the hinges. The magnitude of the horizontal reaction at B is _____ kN (rounded off to 1 decimal place).



3.2.2. General Plane Motion

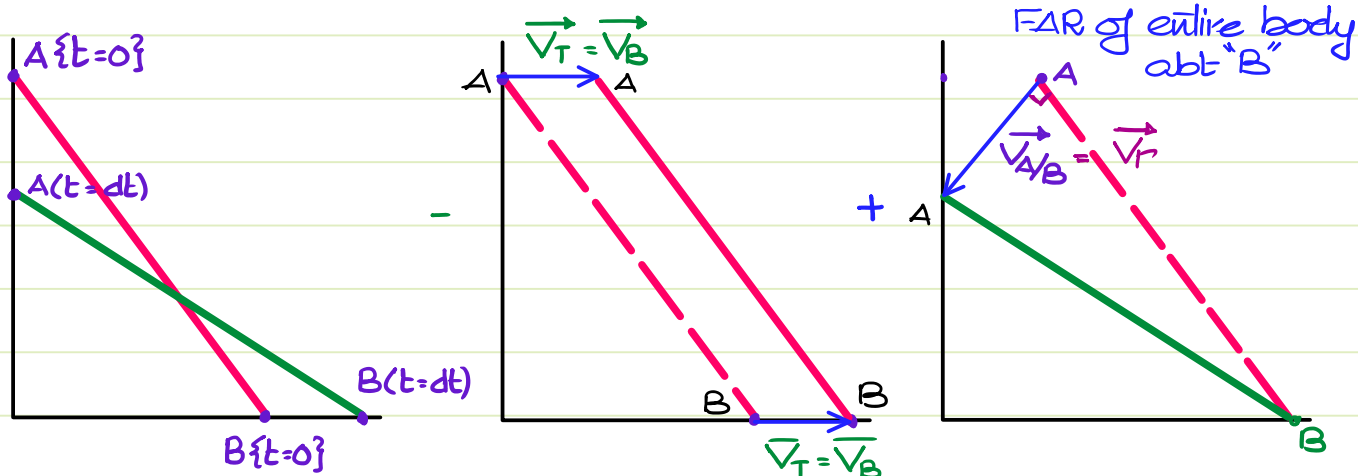
* Introduction

- * GPM = Translation + FAR
- * Pure Rolling / Rolling w/o slip.
- * Instantaneous Centre Method
- * Relative Velocity Method.

General Plane Motion: Motion of a rigid body in a plane such that the body doesn't have any fixed axis. The body may translate throughout the plane. $\therefore$ 2-translatory DOF in mutually $\perp$ axes. The body is also free to rotate, $\therefore$ 1 Rotational DOF about axis $\perp$ to plane. The body has 3 DOF in total.

Examples: Motion of connecting rod in a 4 bar linkage Mech.
Motion of a ladder b/w 2 planes.
Rolling of a body on a surface.

GPM = Translation + Rotation {FAR}:



GPM is a result of 2 simultaneous motions {Translation + Rotation}

Motion 1: The entire body is given translational velocity.

- * The absolute velocity of any particle may be chosen as Translational Velocity {Ex: $\vec{V}_B = \vec{V}_T$ }
- * $\therefore$ Every particle gets same $\vec{V}_T$, $\therefore \vec{V}_T$'s Mag & dirⁿ is same throughout the body.

Motion 2: The particle whose abs. $\vec{V}$ was taken as $\vec{V}_T$ got its absolute motion. $\therefore$ In this stage that particle must be fixed & the entire body must rotate about it. {FAR}

{Ex: FAR abt B}

- * Every particle must have a unique $\vec{V}$ in this stage.
 $\vec{V}_r = r \cdot \omega$. This $\vec{V}_r$ also represents the relative velocity w.r.t the fixed particle {centre of FAR}.

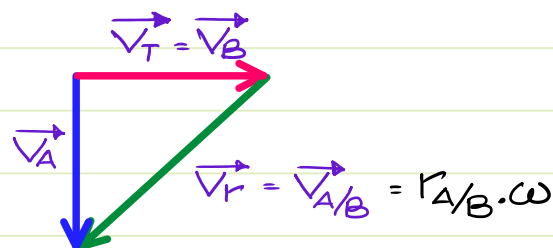
V.V.Imp: The relative velocity of one particle w.r.t another particle in a Rigid body observing GPM must always be $\perp^{\text{ar}}$ to the line joining the particles.

The absolute velocity of any particle is the resultant of the velocities in the two simultaneous motions.

Ex: Motion of A:

$$\vec{V}_A = \vec{V}_T + \vec{V}_r$$

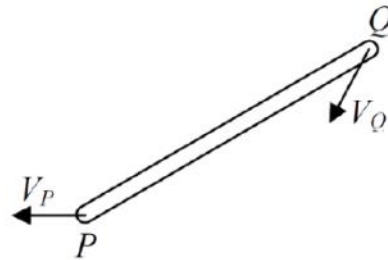
$$\vec{V}_A = \vec{V}_B + \vec{V}_{A/B}$$



Note: If a point having zero instantaneous absolute velocity is chosen for $\vec{V}_T$, then $\vec{V} = \cancel{\vec{V}_T} + \vec{V}_r = \vec{V}_r$. That pt. is known as Instantaneous Centre of Rotation.

IM.

A rigid link PQ is undergoing plane motion as shown in the figure (V_P and V_Q are non-zero). $V_{Q/P}$ is the relative velocity of point Q with respect to point P .



$$\vec{V}_{Q/P} = \vec{V}_Q - \vec{V}_P$$

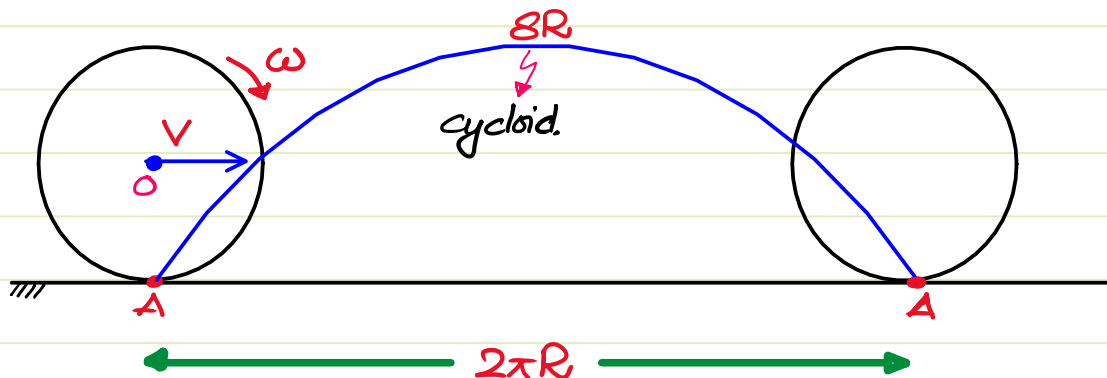
$\therefore \vec{V}_{Q/P}$ must be $\perp$ or $\perp$ to $\vec{PQ}$

Which one of the following is TRUE?

- (A) $V_{Q/P}$ has components along and perpendicular to PQ
- (B) $V_{Q/P}$ has only one component directed from P to Q
- (C) $V_{Q/P}$ has only one component directed from Q to P
- ☒ (D) $V_{Q/P}$ has only one component perpendicular to PQ

Pure Rolling / Rolling w/o Slip: { Slip: Relative motion at pt. of contact? }
 Rolling of a body over a surface such that there is no relative motion at the point of contact { Relative vel. at contact = 0 }
 $\vec{V}_{\text{slip}}$: Relative Velocity at the point of contact.

Rolling w/o slip of a Circular body over a fixed horizontal surf:



displacement of any particle after 1 Revolution: $S = 2\pi R$.
 distance travelled by a surface { peripheral } particle in one complete revolution = SR .
 - trajectory of the peripheral particle: Cycloid.

∴ The Motion is Rolling w/o slip

$$\therefore \vec{v}_{\text{slip}} = \vec{v}_{A/\text{surface}} = 0$$

$$\Rightarrow \vec{v}_A - \vec{v}_{\text{surface}} = 0 \quad \therefore \vec{v}_A = \vec{v}_{\text{surface}}$$

The particle of body in contact with the surface must acquire the $\vec{v}_{\text{surface}}$ in order to avoid slip.

∴ surface is fixed, ∴ $\vec{v}_{\text{surface}} = \vec{v}_A = 0$.

pt. of contact has zero velocity.

The velocity of centre {O} is given by:

$$\begin{aligned} \vec{v}_O &= \cancel{\vec{v}_A} + \vec{v}_{O/A} \\ &= r_{O/A} \cdot \omega \end{aligned}$$

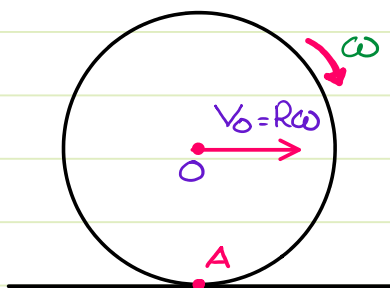
$$\therefore \vec{v}_O = R\omega$$

Note: In pure rolling on fixed surface:

$$\vec{v}_{\text{centre}} = R\omega$$

$$\vec{a}_{\text{centre}} = R\alpha$$

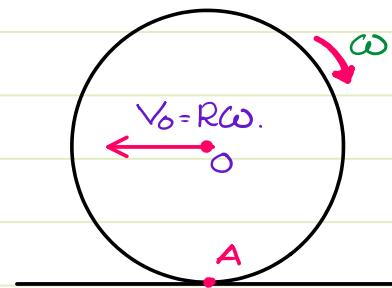
The above mentioned pts are necessary but Not sufficient. If the point of contact has zero velocity, then it is a sufficient condⁿ to say rolling w/o slip.



$$\begin{aligned} \vec{v}_A &= \vec{v}_O + \vec{v}_{A/O} \\ &= (R\omega)\hat{i} + \{R\cdot\omega\}(-\hat{i}) \end{aligned}$$

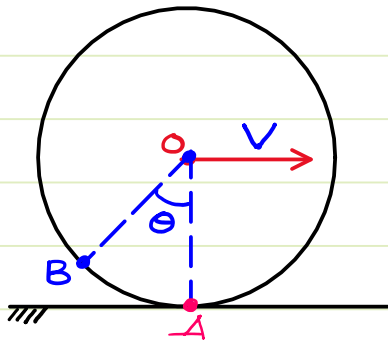
$$\vec{v}_A = 0$$

Rolling w/o slip.



$$\begin{aligned} \vec{v}_A &= \vec{v}_O + \vec{v}_{A/O} \\ &= R\omega\{-\hat{i}\} + R\omega\{-\hat{i}\} \\ &= -2R\omega\hat{i} \neq 0. \end{aligned}$$

∴ Rolling with slip.



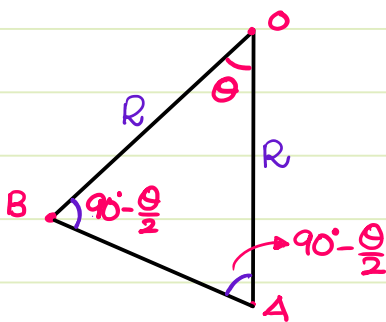
Given: A circular body rolls w/o slip on a stationary surface.
 $\vec{V}_O = R\omega$.

To find: $\vec{V}_B = f(V, \theta)$
 if B is a surface particle & $\angle BOA = \theta$.

Solution: $\because$ Rolling w/o slip on a fixed surface, $\therefore \vec{V}_A = 0$

$$\vec{V}_B = \vec{V}_A + \vec{V}_{B/A} \Rightarrow \vec{V}_B = \vec{V}_{B/A} \quad \text{--- ①}$$

In $\triangle BOA$:



$$\Rightarrow \frac{R_{B/A}}{\sin \theta} = \frac{R}{\sin \{90^\circ - \frac{\theta}{2}\}} \Rightarrow R_{B/A} = \frac{R \cdot \sin(2 \cdot \frac{\theta}{2})}{\cos(\theta/2)}$$

$$\Rightarrow R_{B/A} = \frac{R \cdot 2 \sin(\theta/2) \cdot \cancel{\cos(\theta/2)}}{\cancel{\cos(\theta/2)}}$$

$$\Rightarrow \boxed{R_{B/A} = 2R \sin\left\{\frac{\theta}{2}\right\}}$$

By substituting the value of $R_{B/A}$ in eqⁿ ①

$$V_B = 2R\omega \sin\left\{\frac{\theta}{2}\right\}$$

$$\Rightarrow \boxed{V_B = 2V \sin\left\{\frac{\theta}{2}\right\}}$$

* Using the result of V_B , find the length of cycloid.

$$\begin{aligned} dl &= V_B \cdot dt \\ \Rightarrow dl &= 2R\omega \sin\left\{\frac{\theta}{2}\right\} \cdot dt \\ \theta &= \omega t \\ L &= 2R\omega \int_0^T \sin\left(\frac{\omega t}{2}\right) \cdot dt \end{aligned}$$

$$T = 2\pi/\omega$$

$$L = \frac{2R\omega}{\{\omega/2\}} \cos\left\{\frac{\omega t}{2}\right\} \Bigg|_{t=0}^{t=T=2\pi/\omega}$$

$$L = 4R \cdot \{\cos 0 - \cos \pi\}$$

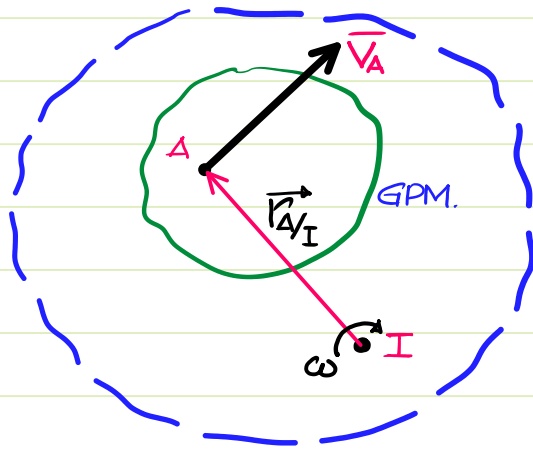
$$\therefore \boxed{L = 8R}$$

*** Instantaneous Centre of Rotation {I Centre Method}

Any body observing GPM {3 DOF} can be imagined to be a subset of a larger imaginary body observing FAR {1 DOF} for a particular instant of time.

⇒ The analysis of GPM body reduces to analysis of FAR body but only for a particular time instant.

The centre of imagined FAR is known as Instantaneous Centre of Rotation {I Centre} $\vec{V}_I = 0$



Imag.
FAR

$$\vec{V}_A = \vec{V}_I + \vec{V}_{A/I}$$

$$\vec{V}_A = \vec{V}_{A/I}$$

$$V_A = r_{A/I} \cdot \omega$$

$\vec{V}_A$ is $\perp^{\text{ar}}$ to $\vec{r}_{A/I}$

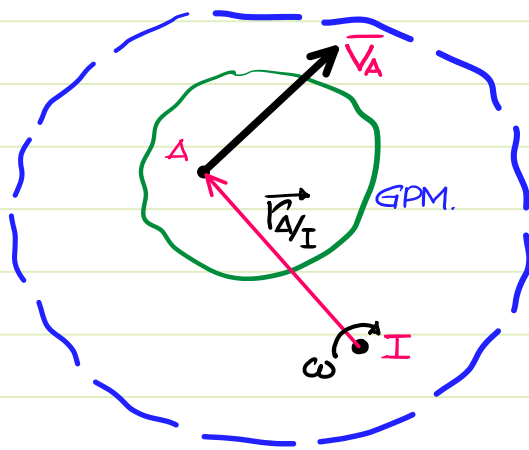
The analysis is only valid for the present instant of time.

*** Instantaneous Centre of Rotation {I Centre Method}

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Imag. FAR

$$\vec{V}_A = \vec{V}_I + \vec{V}_{A/I}$$

$$\vec{V}_A = \vec{V}_{A/I}$$

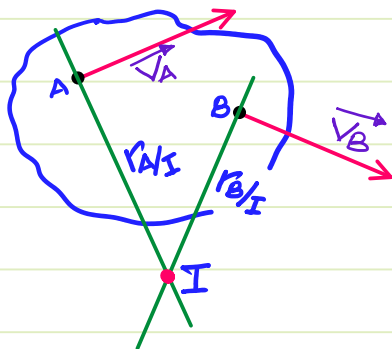
$$V_A = r_{A/I} \cdot \omega$$

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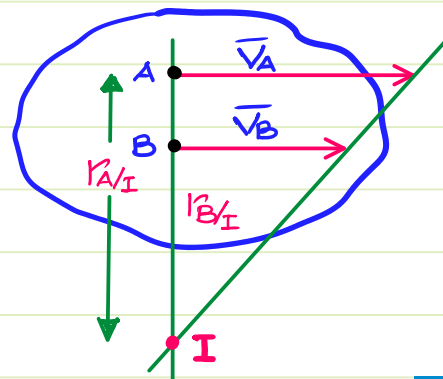
** Location of I Centre {Logic: $\vec{V}$ are $\perp^{\text{ar}}$ to $\vec{r}$ from I Centre}

1. Point of Intersection of $\perp^{\text{ar}}$ to $\vec{V}$:

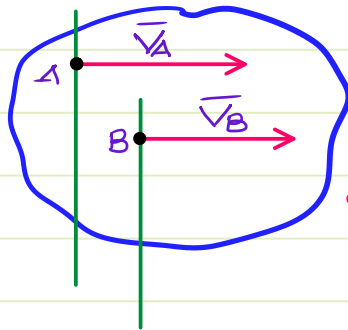


2. If both the particles have common $\perp^{\text{ar}}$

Logic: Ratio of V s = Ratio of radii



3. If the $\perp^{\text{ns}}$ to the $\vec{V}$ do not coincide & they are neither common to both the $\vec{V}$ s, this implies the body is in **pure translation**.
 $\therefore$ the $\vec{V}$ s of all the particles are identical.



Pure Translation

$$\begin{aligned} \therefore \text{I centre is at infinite} &\Rightarrow r \rightarrow \infty \\ &\Rightarrow \omega = 0 \\ \vec{V}_A &= \vec{V}_B \end{aligned}$$

IM.

A circular object of radius r rolls without slipping on a horizontal floor with the centre having velocity V . The velocity at the point of contact between the object and the floor is

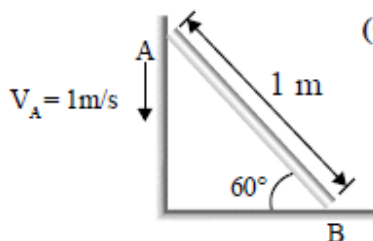
(GATE-14)

- (a) Zero
 (b) V in the direction of motion
 (c) V opposite to the direction of motion
 (d) V vertically upward from the floor.

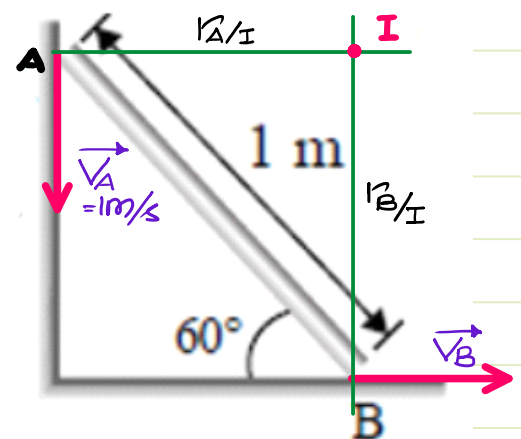
In the case of rolling w/o slip on a stationary surface, I Centre lies at the pt. of contact.

A rod of length 1m is sliding in a corner as shown in figure. At an instant when the rod makes an angle of 60° with the horizontal plane, the velocity of point A on the rod is 1 m/s. The angular velocity of the rod at this instant is

(GATE -1996)



- (a) 2 rad/s
 (b) 1.5 rad/s
 (c) 0.5 rad/s
 (d) 0.75 rad/s



$$\begin{aligned} V_A &= r_{A/I} \cdot \omega \\ \Rightarrow \omega &= V_A / r_{A/I} \\ &= 1 / \cos 60^\circ \end{aligned}$$

$$\omega = 2 \text{ rad/s}$$